



A predictor for dead-time systems based on the Kalman Filter for improved disturbance rejection and robustness

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ABSTRACT

Dead-time processes are common in industry and represent a challenge for feedback control. The use of predictor structures with the controllers can attenuate this. A predictor is proposed here based on the Kalman filter, the Kalman Predictor (KP), that has only one tuning parameter for SISO systems and can be used with non-minimum phase, open-loop unstable and integrative systems of any order, and is also capable of estimating many types of disturbances. It is shown that the KP implicitly defines a Filtered Smith Predictor (FSP), hence, all tools used to analyze the closed-loop properties of the FSP can be used, including robustness and disturbance rejection. Tuning guidelines are provided and simulation examples with comparisons with other predictor structures are used to illustrate the advantages of the proposed KF.

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1. Introduction

The effects of dead-time in closed-loop systems have been a subject of study for many decades now and it still attract the attention of researchers [1,2]. In linear time-invariant systems, if a system has delay the closed-loop characteristic equation is not polynomial anymore in the continuous-time case and, in the discrete-time case, the order of the system increases proportionally to the size of the delay. In both cases, the presence of delay makes the system harder to control with traditional techniques [1].

Nevertheless, dead time remains an essential part of process dynamics modeling, since it can represent naturally occurring phenomena such as information, energy or mass transportation, and it can also be used to approximate high-order systems with low-order ones. Intuitively, the problems caused by dead time in the closed loop are easy to understand: (a) when the input of the process changes, its effect will be felt in the output only after the dead time; (b) disturbances are not felt immediately; (c) because of (a) and (b), the controller tries to correct something that happened in the past [3].

In the seminal work of Smith [4], a structure, which is now called Smith Predictor (SP), was proposed to cope with the effects of dead time. This structure uses information from the input and output of the process to predict the system behavior after the dead time. Then, this signal can be used by a controller, which

can be called primary controller, as if the system did not have dead time. When a predictor such as the SP is used, the dead time does not appear in the closed-loop characteristic equation in the nominal case, greatly simplifying the design of the closed-loop system [3,5]. This also makes the design of both the controller and predictor independent, though the closed-loop dynamics and robustness varies with different tuning and predictor structure.

Throughout the past half-century, many more predictor structures were proposed in the literature. Many of them involve modifications of the original SP [3]. Although the SP solved many problems related to dead-time systems, it could not be applied to open-loop unstable and integrative processes, and had slow disturbance rejection. An important modification of the SP is the Filtered Smith Predictor (FSP) [6], which introduces a filter in the predictor structure that can, if properly tuned, deal with any type of process, and also gives an extra degree of freedom to deal with disturbances and robustness. However, as the order of the system increases its tuning guidelines become more complicated, specially when MIMO systems are considered. New works related to the FSP generally deals with some sort of simplification, or the proposition of tuning guidelines to achieve better disturbance rejection or robustness [7,8].

Another predictor structure is the Generalized Predictor (GP) [9,10] and the Simplified Generalized Predictor (SGP) [11]. One interesting advantage of the SGP is that it intrinsically cope with unstable and integrative processes, hence its tuning guidelines become more simple than in the FSP case. It uses a Finite Impulse Response (FIR) filter and a modified system model to obtain the predictor. There is also the Artstein predictor [12,13], that can

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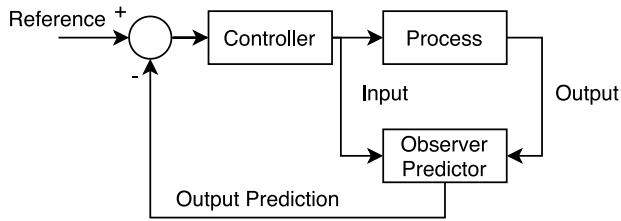


Fig. 1. Observer–predictor structure. Figure based on [3].

cope with unstable dynamics too but its disturbance rejection tends to be slow.

It is also worth noting that many advanced control algorithms implicitly defines a predictor structure. This happens, for instance, with Model Predictive Controllers (MPC), but, as was shown in [1] for the specific case of the Generalized Predictive Control (GPC), the implicit optimal predictor makes the stability margins of the controller dependent on the nominal dead time. This kind of problem is related to the implicit disturbance model and observer.

Regardless of the predictor, as shown in [14], they can all be analyzed through an observer–predictor (O&P) structure (Fig. 1). In the O&P, an observer is used to estimate the plant states and disturbances and, then, a predictor is used to predict the system output d -steps ahead where d is the discrete delay in number of samples. Linear time-invariant observers, such as the steady-state Kalman filter [15] or the Luenberger, are specially interesting because of the separation principle, which makes the design less complex [16].

In [17], the robustness of a dead-time compensation scheme based on the O&P was analyzed for delay, sampling period and plant parameters uncertainties. The authors showed how a stable observer leads to a stable predictor using an augmented model, and they also used a robust Linear Matrix Inequalities (LMI) based controller on a quadrotor. A robust H^2 controller is designed in [14]. An O&P based on the Kalman filter is used in [18] to control a quadrotor, but robustness properties were not analyzed. On all these works, frequency domain analysis is overlook in favor of an LMI framework to design and analyze the closed loop.

In [19], an O&P is proposed where the observer is tuned with pole-placement method and a secondary frequency-domain filter is used to adjust disturbance rejection and robustness. It considers step and sinusoid disturbances in the estimation. Although the authors analyze the output error bounds and closed-loop stability, the effects of the predictor on closed-loop robustness are not addressed.

In [20], a predictor scheme is proposed using a nonlinear version of the Kalman Filter (KF), the Extended Kalman filter (EKF), but no disturbance is estimated, making the predictor unable generate an offset-free prediction of the system states or output, which impacts disturbance rejection. The authors improve this predictor in [21], where an Extended State Observer (ESO) is used to also estimate disturbance effects. However, the analysis of the predictor properties is not complete, specially in regards to its tuning parameters and its effects on the closed-loop properties.

It is also worth mentioning that, even though the KP proposed here considers a fixed delay in its formulation, it can still deal with a time-varying delay. This kind of delay appears naturally in, for example, some mass transport phenomena in hydraulics systems [22], and in some chemical reactors [23], and has been receiving increasing interest from academia. In some works, the authors propose control schemes to deal with this type of delay. For instance, in [24], an Internal Model Control (IMC) based formulation was used to control a nonlinear static process

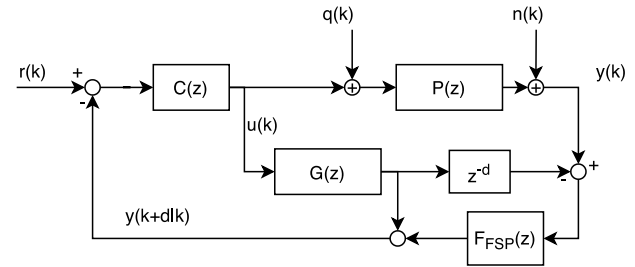


Fig. 2. Filtered Smith Predictor (FSP) block diagram.

under model uncertainties, including a variable delay. In [25], a novel observer–predictor control is proposed to stabilize uncertain systems subject to unknown but bounded arbitrarily-fast time-varying input and output delays. Nonetheless, in [26–31], the authors, using different methods, showed that a predictor with fixed delay can still be robust even with a time-varying delay. This property was demonstrated with a Lyapunov–Krasovskii functional in [26], a Linear Matrix Inequalities (LMI) approach in [27,31], and a modified Nyquist Stability Criterion that took into account time-varying delay in [28]. This last work was extended in [29] to also account for other model uncertainties. Moreover, a model uncertainty in a receptance-based stability criteria for systems with time-varying delay was proposed in [30] along with a receptance-based IMC design approach. Thus, these works show that fixed delay predictors are still relevant.

Hence, in this work, the authors propose a modification of the Kalman Predictor (KP), which is a Kalman filter based O&P with fixed delay which simplifies the one in [32]. Some characteristics of the proposed KP are:

- supports to the estimation of different disturbance types similarly to [19,21], which confers interesting disturbance rejection and robustness capabilities;
- it inherently copes with systems of any order and dynamics, such as non-minimum phase and unstable or integrative poles without any additional procedure;
- it can be used in state–space or transfer-function form;
- it has only one tuning parameter that can be adjusted for better disturbance rejection or robustness;
- the predictor is defined directly in discrete time, making the implementation straight-forward.

Some of these characteristics were present in other works cited above but, to the best of our knowledge, none of them had all of these advantages. It is also worth noting that the KP implicitly defines an FSP, whose properties are well known in the literature. Hence, all the tools used to interpret the closed-loop characteristics of a dead-time compensation scheme with an FSP can be used for the KP, specially the frequency-domain ones. This work will focus only on SISO systems and the MIMO case will be addressed in future works. Note that in MIMO processes, when multiple dead-times are considered, the predictor structure is more complex [33].

The remaining of this work is divided in the following way: preliminaries and notation are presented in the next subsection. The KP is introduced in Section 2. The frequency domain analysis and tuning guidelines of the KP are in given in Section 3. In Sections 4 and 5 some simulation results and the conclusions, respectively, are given.

1.1. Preliminaries: Filtered Smith Predictor

The Filtered Smith Predictor (FSP) properties will be briefly revised here. It is essential to understand them in order to fully

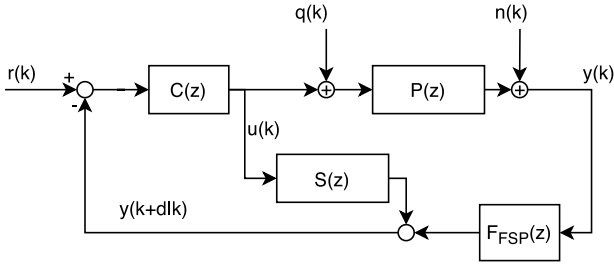


Fig. 3. Implementation structure of the FSP for unstable and integrative systems.

comprehend the advantages of the proposed Kalman Predictor. The FSP, shown in Fig. 2 uses a dead-time free model of the process $G(z)$ to compute the predicted value of the output after the delay. This predicted value is sent via feedback to a primary controller $C(z)$. However, this prediction must be corrected in some way to take into account disturbances and model mismatch. In the FSP, this is done by calculating the prediction error, the difference between the real output of the process and the nominal output of the process, which is then filtered by $F_{FSP}(z)$, and finally added to the nominal prediction. Also, in Fig. 2, $q(k)$ is an input disturbance, $n(k)$ is an output disturbance, or noise, $r(k)$ is the reference, $u(k)$, $y(k)$, $y(k+d|k)$ are the process input, output, and output prediction after the delay, respectively. The FSP characteristics are listed below [1]:

- Dead-time compensation, i.e., the characteristic equation of the closed-loop transfer functions are not influenced by the dead time in the nominal case:

$$\frac{Y(z)}{R(z)} = \frac{C(z)G(z)}{1 + C(z)G(z)} z^{-d}, \quad (1)$$

$$\frac{Y(z)}{Q(z)} = P(z) \left(1 - \frac{C(z)G(z)z^{-d}}{1 + C(z)G(z)} F_{FSP}(z) \right), \quad (2)$$

$$\frac{Y(z)}{N(z)} = \left(1 - \frac{C(z)G(z)z^{-d}}{1 + C(z)G(z)} F_{FSP}(z) \right). \quad (3)$$

Hence, the primary controller $C(s)$ can be tuned disregarding the dead-time of the process, simplifying the design procedure.

- The closed-loop set-point tracking is not affected by the prediction-error filter $F_{FSP}(z)$ in the nominal case.
- The closed-loop disturbance rejection response can be adjusted through the prediction-error filter $F_{FSP}(z)$. Depending on the filter properties, this response can be accelerated or slowed down.
- For unstable and integrative systems, the open-loop poles of the process in (2) must be removed to achieve stability and zero-error disturbance rejection. To do this, the following condition must be satisfied for every undesirable p_i open-loop pole of $P(z)$,¹

$$(1 - z^{-d} F_{FSP}(z))|_{z=p_i} = 0, \quad \forall i. \quad (4)$$

- For unstable and integrative processes, to guarantee the internal stability of the predictor, the FSP must be implemented with the structure shown in Fig. 3, where

$$S_{FSP}(z) = G(z)(1 - z^{-d} F_{FSP}(z)), \quad (5)$$

and $S(z)$ should be a stable transfer function for any $G(z)$.

¹ This condition changes in the particular case of integrative processes or multiple roots in the same position [1] though the core idea remains the same.

- Given a model mismatch that is represented by the multiplicative error of a family of models in the frequency domain $\delta P_i(z)$, with $P_i(z) = P(z)(1 + \delta P_i(z))$, the closed-loop system remains stable for all $P_i(z)$ if the following robustness condition, which is based on the Nyquist stability criterion, is true:

$$\left| \frac{C(z)G(z)}{1 + C(z)G(z)} z^{-d} \right| |F_{FSP}(z)| < \frac{1}{|\delta P_i(z)|}, \quad \forall z = e^{\pm j\omega T_s}, \omega \in [0, \pi/T_s], \forall i. \quad (6)$$

Hence, the closed-loop system can become more robust if $F_{FSP}(z)$ has low-pass characteristics, and vice-versa. For a time-varying delay, this condition changes to the one below [29]

$$\left| \frac{C(z)G(z)}{1 + C(z)G(z)} z^{-d_n} \right| |F_{FSP}(z)| < \frac{1}{|\delta P_i(z)| + (1 + |\delta P_i(z)|)\tilde{N}|1 - z|}, \quad \forall z = e^{\pm j\omega T_s}, \omega \in [0, \pi/T_s], \forall i, \quad (7)$$

where the delay is now given by $d(k) = d_n + \delta d(k)$, with d_n and $\delta d(k)$ being, respectively, the time-invariant and time-variant part of the delay. Also, $0 \leq d(k) \leq NT_s$, with $N \in \mathbb{R}^+$, $\tilde{N} = \sqrt{[N]^2 - 2g[N] + g}$ and $g = N - [N]$. Notice that if $N = 0$, i.e., no time-varying delay, (7) becomes (6).

1.2. Preliminaries: system model and notation

The following discrete SISO state-space model will be used to represent the system:

$$\begin{cases} x(k+1) = Ax(k) + Bu(k-d) + B_\phi \phi(k) \\ y(k) = Cx(k) \end{cases} \quad (8a)$$

$$y(k) = Cx(k) \quad (8b)$$

with $x \in \mathbb{R}^n$ for the states, $u \in \mathbb{R}$ for the control action, $\phi \in \mathbb{R}^q$ for the disturbance and $y \in \mathbb{R}$ for the measured output. A unique known input delay d is considered. Note that for physical systems there is always a one sample delay between the process' inputs and its effect on the output, hence, there is no matrix D in the previous model, as is common for general state-space representation. In the frequency domain, using the Z-Transform, this model is given by

$$Y(z) = P(z)U(z) + P_\phi(z)\phi(z), \quad (9)$$

where $P(z) = G(z)z^{-d} = C(zI - A)^{-1}Bz^{-d}$, and $P_\phi(z) = G_\phi(z)z^{-d} = C(zI - A)^{-1}B_\phi$.

Notation. $x(k+j|k)$ refers to the prediction of x at time $k+j$ given information up to time k .

2. The Kalman predictor

The idea behind the Kalman Predictor is to use an observer, which in this case is the Kalman filter, to estimate the current state of the system and then use the system model to compute the state d -steps ahead. By doing this, a Dead-Time Compensation (DTC) scheme is generated which is akin to the FSP.

The KP algorithm is simple and was introduced in [32]:

1. Use the Kalman Filter to estimate the current state of the process taking into account possible disturbances.
2. Obtain the prediction of the system after the dead time by integrating the system model.
3. Use this state prediction in the feedback controller. Or, if it is an output feedback controller, use the output prediction.

4. Repeat after each sampling time.

In this work, some modifications will be introduced in relation to [32] that will make the KP more general by making it able to cope with different types of disturbances, and simpler to be tuned. One interesting observation is that any state observer could be used instead of the KF. However, the KF has some advantages: i) it is not necessary to think in terms of eigenvalues/poles of the observer; ii) the KF automatically takes into account the system dynamics (unstable, non-minimum phase, etc.) [34].

2.1. STEP 1: State-estimation

Firstly, the user need to obtain a state-space model of the system in consideration, even if the intention is to use a transfer function implementation. If this is the case, it does not matter if the states represent physical variables or not, the state-space model will only be used as an intermediary step to obtain the predictor.

Then, assuming that the disturbances present in the system (8) are unknown, and that they can represent both model mismatch and external signals, their estimation is essential to obtain an offset-free predictions of the system states (or output). Each particular problem might have different types of disturbances but, in general, it is reasonable to assume that the disturbance is constant [35]. But in some problems a sinusoid, or ramp dynamics, might represent the disturbance better and, in some cases, a combination of types might be necessary.

Hence, the first step is to properly identify which types of disturbances the predictor will have to consider. Then, the user has to create an augmented state-space representation of the system where the disturbances are part of the states of this new system. For example, consider the case where a constant disturbance and a sinusoid one are present, they can be represented by the following difference equations, respectively:

$$\phi_1(k+1) - \phi_1(k) = 0 \quad (10)$$

$$\phi_\omega(k+2) - 2\cos(\omega T_s)\phi_\omega(k+1) + \phi_\omega(k) = 0 \quad (11)$$

where T_s is the sampling time and ω the disturbance frequency.

The Kalman filter, which will be used as an observer, has a stochastic formulation in which zero-mean additive Gaussian noises are considered in the states ($w \in \mathbb{R}^{n_w}$) and output ($v \in \mathbb{R}$). They are defined by diagonal covariance matrices $\mathcal{Q}$ and $\mathcal{R}$, respectively, which means that the noises are uncorrelated. Then, the augmented model can be written, for this particular case, as

$$\begin{cases} \bar{x}(k+1) = \bar{A}\bar{x}(k) + \bar{B}u(k-d) + Hw \\ y(k) = \bar{C}\bar{x}(k) + v \end{cases} \quad (12a)$$

$$\text{with } \bar{x}(k)^T = [x(k)^T, \phi_1(k), \phi_2(k)^T]^T, C_{\phi_1} = 1, C_{\phi_2} = [1, 0], \phi_2(k)^T = [\phi_\omega(k), \phi_\omega(k-1)]^T,$$

$$\bar{A} = \begin{bmatrix} A & B_\phi C_{\phi_1} & B_\phi C_{\phi_2} \\ 0 & 1 & 0 \\ 0 & 0 & A_\omega \end{bmatrix}, \quad A_\omega = \begin{bmatrix} 2\cos(\omega T_s) & -1 \\ 1 & 0 \end{bmatrix}$$

$$\bar{B} = \begin{bmatrix} B \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \bar{C} = [C, 0, 0, 0]$$

and $H \in \mathbb{R}^{(n+n_q) \times n_w}$, with n_q being the number of disturbance states.

For the proposed KP algorithm, the disturbances affect the system through the input, i.e., $B_\phi = B$, this is done because disturbances due to model mismatch are dependent on control input, hence they are better represented with this consideration.

The matrix H selects where the noise w appears in the state equations. In [32], an assumption was made that there was an

individual noise signal w_i for each state of (12), i.e., $n_w = n + q$ for SISO systems, which is the standard approach when designing a Kalman filter as a state observer. However, in this paper, we propose that the number of noise signals will always be one, independent of the number of states or types of disturbances considered in the augmented model. In this simplified case, the interest lies only in the system output estimation after the dead time, and not on the state estimation, hence, the noise in the states can be removed. Also, by using only one noise signal, the tuning becomes considerably easier, because the number of tuning parameters of the KF is directly related to the number of noise signals v and w . For SISO systems, there will only be two tuning parameters, $\mathcal{Q}$ and $\mathcal{R}$, which are scalar numbers in this case.² In the example, H is then given by

$$H = \begin{bmatrix} 0_{n \times 1} \\ 1 \\ 1 \\ 0 \end{bmatrix} \quad (13)$$

where 0_j is a zeros matrix with dimension j .

It is important to notice that the user can follow the procedure explained here to estimate any type and number of disturbances. The only restriction is that the system (12) must be detectable. The detectability of pair $(\bar{A}, \bar{C})$ guarantees a steady-state Kalman filter with gain L that is marginally stable ([36], Theorem 26).

With the augmented system, the user can use any software tool to compute the steady-state (fixed gain) Kalman filter to estimate the system states. These tools generally return $L, M \in \mathbb{R}^{(n+q) \times 1}$ which are, respectively the filter gain and the innovation gain, with $L = \bar{A}M$.

The procedure explained above is done off-line in a design phase. The first on-line step of the algorithm is to use the steady-state KF to estimate the augmented state $\bar{x}(k|k)$ [36], which has the following dynamics:

$$\begin{cases} \bar{x}(k+1|k) = (\bar{A} - L\bar{C})\bar{x}(k|k-1) + [L \quad \bar{B}] \begin{bmatrix} y(k) \\ u(k-d) \end{bmatrix} \\ \bar{x}(k|k) = (I - M\bar{C})\bar{x}(k|k-1) + [M \quad 0] \begin{bmatrix} y(k) \\ u(k-d) \end{bmatrix} \end{cases} \quad (14a)$$

Eqs. (14)(a) and (14)(b) represent, respectively, the filter dynamics and its output. To obtain the system output prediction $y(k|k)$, multiply (14)(b) by $\bar{C}$.

2.2. STEP 2: Prediction

As a result of STEP 1, the estimated state $\bar{x}(k|k)$ is now known. So, it is necessary to obtain the predicted state after the dead time at $k+d$ as a function of $\bar{x}(k|k)$, which is done by integrating (14) d steps in the future. As shown in [32], it is possible to combine the estimation of the Kalman filter with the d -steps ahead prediction in a single dynamic system. The Kalman Predictor is then given by

$$\begin{cases} \bar{x}(k+1|k) = (\bar{A} - L\bar{C})\bar{x}(k|k-1) \\ \quad + [L \quad 0 \quad \dots \quad 0 \quad \bar{B}] \bar{u}(k) \\ y(k+d|k) = [\bar{C}\bar{A}^d (I - M\bar{C})]\bar{x}(k|k-1) \\ \quad + \bar{C} [\bar{A}^d M \quad \bar{B} \quad \bar{A}\bar{B} \quad \dots \quad \bar{A}^{d-1}\bar{B}] \bar{u}(k) \end{cases} \quad (15a)$$

$$\text{with } \bar{u}(k) = [y(k)^T, u(k-1)^T, \dots, u(k-d)^T]^T.$$

² It is worth mentioning that even if the user chooses to use one noise signal per state, the KP will still work as described, the only difference is that its tuning will be harder.

Although in this work the system output prediction after the delay is being used, the KP can easily be adapted to output all state prediction after the dead time $\bar{x}(k+d|k)$, if a state-feedback controller is used. As will be seen in Section 3, the analysis becomes easier when the output prediction is considered. Also note that the inputs of this system are the system output and delayed inputs. Hence, it is possible to apply the Z-transform to obtain the transfer function for implementation.

To finalize the algorithm, execute steps 3 and 4, which are trivial.

3. Frequency domain analysis

As stated in Section 1.1, the FSP has many interesting properties that can be analyzed in the frequency domain [1]. This section shows how the KP implicitly defines an FSP and, hence, the analysis tools for FSP can be used for the KP.

In the SISO case, the KP (15) has the prediction $y(k+d|k)$ as output, and two inputs: (i) $u(k)$, the manipulated variable, and its delayed values; and (ii) $y(k)$, the measured output. Using the Z-Transform in (15):

$$Z\{y(k+d|k)\} = Y_d(z) = F_{KP}(z)Y(z) + S_{KP}(z)U(z), \quad (16)$$

with

$$F_{KP} = \bar{C} \left[\bar{A}^d (I - M\bar{C}) (zI - \bar{A} + L\bar{C})^{-1} L + \bar{A}^d M \right],$$

$$S_{KP} = \bar{C} \left[\bar{A}^d (I - M\bar{C}) (zI - \bar{A} + L\bar{C})^{-1} \bar{B}z^{-d} + \sum_{j=1}^d \bar{A}^{j-1} \bar{B}z^{-j} \right].$$

This last equation indicates that the inputs of the KP, $Y(z)$ and $U(z)$, are filtered by $F_{KP}(z)$ and $S_{KP}(z)$, respectively, to generate the estimated output after the dead-time.

From Fig. 3, it can be easily shown that the output prediction of the FSP can be represented in the frequency domain by

$$Y_d(z) = F_{FSP}(z)Y(z) + S_{FSP}(z)U(z). \quad (17)$$

with $S_{FSP}(z)$ as defined in (5).

Comparing (16) and the FSP prediction equation (17), it is clear that the KP at least have some similarities with the FSP. They both have the same inputs, which are then filtered to generate the output prediction after the dead-time. As shown in [32], this is more than a superficial similarity.

Theorem 1. *The Kalman Predictor (KP), defined by (15) and (16), implicitly defines a Filtered Smith Predictor (FSP) (described by (17)) where the prediction-error filter $F_{FSP}(z)$ is equal to $F_{KP}(z)$ from (16).*

Proof. See [32].

This theorem makes the closed-loop analysis of the KP in the frequency domain much more simple. Since the FSP and the KP are equivalent, all the properties of the FSP given in Section 1.1 are valid for the KP. All the user needs to do is to obtain $F_{KP}(z)$, the implicit prediction-error filter of the KP given by (16).

One interesting fact is that the FSP allows the correction of one important flaw of the original Smith Predictor. The SP cannot be used with integrative or unstable systems. This happened for two reasons, internal stability of the predictor and unstable closed-loop (in the unstable case). The latter can easily be checked through (2). Notice that the transfer function has the system model $P(z)$ as a factor, hence, if the system is open-loop unstable, (2) is also unstable. Regarding the settling time of the disturbance rejection, this poses a problem even if the system is stable because the response will always have its open-loop poles.

Also, note that $S_{FSP}(z)$, given by (5), has the system model as a factor, causing the same instability problem of (2). However, in [1], the authors showed that if $F_{FSP}(z)$ is properly tuned, it is possible to remove the undesired poles from $S_{FSP}(z)$, solving this particular flaw of the SP. The condition that must be satisfied for this to happen is (4), which will make $S_{PSF}(z)$ have zeros at the same place of the undesired poles, forcing a cancellation, thus eliminating the undesired dynamics.

The KP, however, inherently does this independent of tuning. This can be shown with the following corollary:

Corollary 1. *Given the detectability of system (12), Theorem 1 and the KP defined by (15) and (16), independent of tuning, the resulting implicit $S_{KP}(z)$ will never have any open-loop poles of $P(z)$, i.e.,*

$$(1 - z^{-d}F_{KP}(z))|_{z=p_i} = 0, \quad \forall i, \quad (18)$$

where p_i represents the i th pole of $P(z)$.

Proof. From (16) notice that only the poles of $F_{KP}(z)$, that come from the term $(zI - \bar{A} + L\bar{C})^{-1}$, appear in $S_{KP}(z)$. But, by (5), the open-loop poles, given by $(zI - \bar{A})^{-1}$, should also appear in $S_{KP}(z)$. Since they do not, by Theorem 1, this would only happen if (18) is true, or if $L = 0$, which cannot happen, otherwise the observer would be open-loop. $\square$

Since the Kalman Predictor removes all open-loop poles of $S_{KP}(z)$, the problems described earlier do not happen. $S_{KP}(z)$ is always stable and the open-loop poles of $P(z)$ do not appear in disturbance rejection responses, given by (2) and (3). An intuitive example that shows this is an unstable case. Considering that the augmented system is detectable, the Kalman estimator is stable [34], i.e., although the system model is used, the unstable poles are removed from the observer.

Hence, one great advantage of the proposed Kalman Predictor is that it can be used for any input-time-delay process of any order, with no need to design specific guidelines for unstable, integrative or non-minimum phase processes as some of the ones studied in the literature [9,37–40].

3.1. Tuning guidelines of the KP

The proposed KP has two tuning parameters, the variance matrices $\mathcal{Q}$ and $\mathcal{R}$ of the disturbance state and of the output, respectively. Changing their values will affect the gains of the Kalman Filter which, in turn, modify the prediction-error filter $F_{KP}(z)$. The relation between the filter characteristics, such as bandwidth, maximum peak, etc., and the KP tuning parameters are hard to obtain algebraically, because the relation between the system model (gain, poles, and zeros positions, and delay), and the tuning parameters are nonlinear. However, one interesting characteristic was detected that helps the tuning procedure.

Before discussing the proposed guideline, the frequency characteristics of the $F_{KP}(z)$ will be shown for different systems and tuning parameters. In [41], the authors give a list of interesting systems to benchmark PID controllers, which will be partially used here. Consider the following systems:

$$P_1(s) = \frac{1}{(s+1)^3} e^{-5s}, \quad T_s = 0.1 \quad (19)$$

$$P_2(s) = \frac{1}{(s+1)(1+\alpha s)(1+\alpha^2 s)(1+\alpha^3 s)} e^{-3s},$$

$$T_s = 0.1, \alpha = 0.5 \quad (20)$$

$$P_3(s) = \frac{1-\alpha s}{(s+1)^3} e^{-2s}, \quad T_s = 0.1, \alpha = 0.1 \quad (21)$$

$$P_4(s) = \frac{100}{(s+10)^2} \left(\frac{1}{s+1} + \frac{0.5}{s+0.05} \right) e^{-10s}, \quad T_s = 2 \quad (22)$$

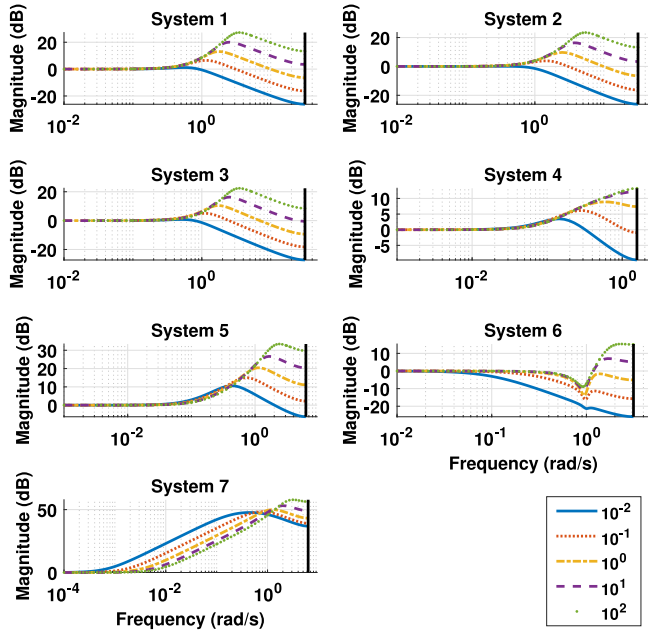


Fig. 4. Frequency domain magnitude plot of filter $F_{KP}(z)$ for varying values of Q and different systems for integral disturbance.

$$P_5(s) = \frac{(s+6)^2}{s(s+1)^2(s+36)} e^{-7s}, \quad T_s = 0.5 \quad (23)$$

$$P_6(s) = \frac{\omega_0^2}{(s+1)(s^2 + 2\zeta\omega_0s + \omega_0^2)} e^{-15s}, \quad T_s = 1, \omega_0 = 1, \zeta = 0.1 \quad (24)$$

$$P_7(s) = \frac{1}{s^2 - 1} e^{-5s}, \quad T_s = 0.5 \quad (25)$$

Note that each system has different poles and zeros which cover most types of system characteristics: non-minimum phase, integrative, stable, unstable, multiple poles, high order dynamics, and oscillatory dynamics. Their discrete state-space models were obtained considering the respective sampling time.

To make the analysis simpler, the output variance matrix $\mathcal{R}$ will be set to 1 for all cases. This is done because the variances values are relative to each other for the steady-state KF and a SISO system, i.e., if all variances are multiplied by a fixed number the result is the same.

In Fig. 4 the bode-plot of $F_{KP}(z)$ for different values of Q are shown for each system considering only an integral disturbance model. Notice that as the variance of the disturbance-state increases the magnitude response of the filter at medium and high frequencies also increases for all systems. This indicates that $F_{KP}(z)$ will have faster poles which, in turn, indicates a faster disturbance rejection since the closed-loop disturbance rejection response (2) is dependent on $F_{KP}(z)$. On the other hand, the closed-loop robustness will diminish according to the Nyquist stability criterion (6), since higher bandwidth and magnitude makes it easier to violate the criterion for a certain model mismatch, which is a natural trade-off.

A more intuitive interpretation for this is that, when the variance Q has higher values, the KP must try to extract more information from the measured output to properly estimate the disturbance state, which is varying too much, and vice-versa, i.e., if the variance is low, less information is needed for the estimation [32]. This will make the predictor converge more rapidly, but will make it less robust.

An interesting case is System 7 (25), which has unstable poles. The frequency characteristics have high amplification at medium and high frequencies independently of Q . This happens because the predictor must guarantee a certain feedback speed to cope with the unstable dynamics of the process. This behavior is well documented for the FSP applied to unstable systems [1].

Taking this into account, the following guideline can be used:

For SISO systems and $\mathcal{R} = 1$, the higher the value of the variance of the disturbance state Q , the higher the bandwidth of $F_{KP}(z)$ and its infinity norm $\|F_{KP}(z)\|_\infty$, $\forall z = e^{\pm j\omega T_s}$, $\omega \in [0, \pi/T_s]$. Hence, higher values of Q will help with disturbance rejection, whereas lower values will increase robustness, which is given by (6). For stable and unstable processes, a good initial value is a weight of 1 normalized by the process gain $K_e = P(1)$: $Q = 1/K_e^2$. If the process is integrative, use K_v , the velocity gain.

4. Illustrative examples

In this section, some illustrative examples will be used to present the applicability of the Kalman Predictor. In all cases, the primary controller will be the same, that is, only the predictor will be changed, which is the focus of this work. However, ideally, the designs of the primary controller and of the predictor are done in parallel to achieve adequate tracking and robustness capabilities, as in [6].

4.1. Example 1: Stable case with sinusoid disturbances

Consider the following dead-time system:

$$P(s) = \frac{1}{3s+1} e^{-2s}. \quad (26)$$

The primary controller was designed considering the delay-free nominal model, aiming to achieve an offset free steady-state for step references and sinusoid disturbances with frequency $\omega = 0.7$ [rad/s], and also have a nominal closed-loop settling time of 4.5 [s], without the delay, for reference step changes. This resulted in the following controller:

$$C(s) = 4.3423 \frac{(s+1)(s+0.5)^2}{s(s^2+0.49)} \quad (27)$$

The controller is discretized using the Tustin method with sampling time $T_s = 0.1$ [s].

For the predictor structure, two KPs will be used, one that considers only a constant disturbance in the model, and another that considers a step and sinusoid disturbance with frequency $\omega = 0.7$ [rad/s]. In both cases, $Q = 1$ was used, resulting in the following prediction error filters:

$$F_{KP1} = \frac{0.53871z(z-0.9457)}{z^2 - 1.741z + 0.7705}, \quad (28)$$

$$F_{KP2} = \frac{4.9238z(z-0.9929)(z^2 - 1.93z + 0.9329)}{(z-0.9951)(z-0.7377)(z^2 - 1.619z + 0.7105)}. \quad (29)$$

Notice that in the second case the filter is more complex because of the additional dynamics imposed by the sinusoid dynamics inserted in the augmented system of the Kalman Filter.

In Fig. 5, the closed-loop responses are presented. A reference change from 0 to 1 is made at $t = 5$ [s], and an input sinusoid disturbance with frequency $\omega = 0.7$ [rad/s] and 0.5 amplitude is added at $t = 20$ [s].

For step reference changes, both cases have the exact same result. However, they have different behaviors in the presence of sinusoid disturbances. Even though the primary controller $C(s)$ is the same, and one that was designed to reject sinusoid disturbances, only the KP2 case achieves this specification. This happens because $y(k+d|k)$ is incorrectly estimated by KP1,

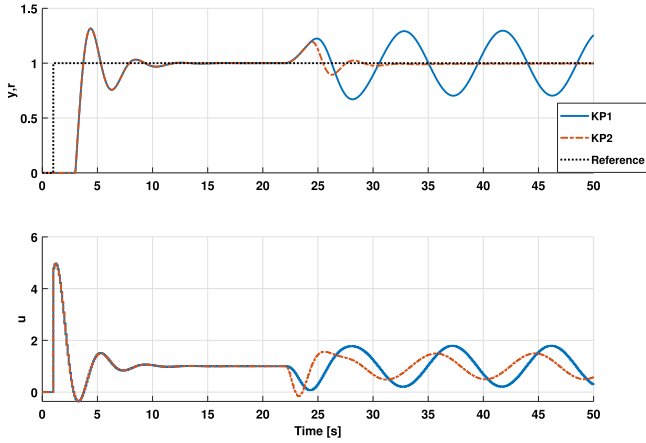


Fig. 5. Example 1: closed-loop responses.

which is incapable of correct estimation in the presence of disturbances that are not of the constant kind. Hence, the controller makes the wrong action to correct the disturbance, resulting in a steady-state error. This illustrates the importance of including the expected disturbance dynamics in the predictor.

Remark 1. It is worth mentioning that the primary controller does not have to necessarily include the disturbance dynamics to achieve disturbance rejection as long as the predictor can properly estimate the disturbance and guarantee that the closed-loop disturbance transfer functions (2) and (3) have zero gain at the disturbances frequencies. This, however, changes the conditions that the predictor must satisfy [1]. This approach will be investigated in future works.

4.2. Example 2: Integrating system with time-varying delay

This example will be used to compare the proposed Kalman Predictor with two other strategies: the Filtered Smith Predictor with tuning guidelines from [6]; and the Simplified Generalized Predictor (SGP) [11]. Keep in mind that the predictor structures that will be shown here are interchangeable, i.e., with proper tuning they can yield the same results since they implicitly define the same FSP structure. However, this can be hard to achieve considering the tuning guidelines suggested in the works where they were presented. These structures will also be tested against a time-varying delay to show that, even though they consider fixed delays in their tuning procedures, they still manage to confer a degree of robustness against this type of delay. This is a common situation in industrial practice, when a predictor-based controller is tuned for a robust behavior of a plant with a delay in the interval $[d_{min}, d_{max}]$. In most cases, robustness is also obtained for a varying delay in the same interval.

Consider the following model which represents a chain of evaporators (tanks) usually found in the food industry [42]:

$$P_n(s) = \frac{-0.1}{s(5s+1)} e^{-5}. \quad (30)$$

In this example, time constants and dead times are in minutes and the output level and input flow are in [%], and they represent the controlled and manipulated variables, respectively.

As in the previous case, a primary controller and a reference filter were designed to obtain a nominal closed-loop settling time of approximately 30 [min] with at most 5 [%] peak for step reference changes:

$$C(s) = \frac{-3.9872(z - 0.9275)(z - 0.8187)}{(z - 1)(z - 0.7)}, \quad (31)$$

$$F(s) = \frac{0.453(z - 0.84)}{(z - 0.9275)}, \quad (32)$$

with $T_s = 1$ [min].

The predictors were tuned to obtain a disturbance rejection with approximately the same settling time. For the FSP,

$$F_{FSP}(z) = \frac{0.5421(z - 0.9592)(z - 0.897)(z^2 - 1.766z + 0.8166)}{(z - 0.8)^4(z - 0.9275)}. \quad (33)$$

For the SGP, the tuning parameters were $\lambda = 0.85$, $m = 1$ and $n_h = 2$, resulting in the following filters

$$\begin{aligned} S_{SGP,num}(z) &= -0.0093654(z + 0.9355)(z - 1) \\ &\quad (z^2 + 0.4933z + 0.1459) \\ &\quad (z^2 - 1.075z + 0.4433), \\ S_{SGP,den}(z) &= z^3(z - 0.85)^4, \end{aligned} \quad (34)$$

$$F_{SGP}(z) = \frac{0.067462(z - 0.9586)(z - 0.8188)z^2}{(z - 0.85)^4}. \quad (35)$$

Finally, for the Kalman Predictor, the tuning parameter was $Q = 0.01$, resulting in the following equivalent FSP filter:

$$F_{KP1}(z) = \frac{0.1726z(z - 0.9454)(z - 0.8204)}{(z - 0.8142)(z^2 - 1.868z + 0.877)}. \quad (36)$$

Also, to show the improved disturbance rejection, a second KP was tuned with $Q = 1$ to decrease the settling time of the response to input disturbances.

$$F_{KP2}(z) = \frac{0.6711z^3 - 1.145z^2 + 0.4887z}{z^3 - 2.432z^2 + 2.002z - 0.5548}. \quad (37)$$

In this particular case, the advantages of the KP become more evident. Since the order and type of the system is higher than in the previous case, the tuning guidelines of the FSP and of the SGP change slightly, whereas nothing changes for the KP. Specially for the FSP, for higher order system, or if the system has special dynamics (non-minimum phase, and unstable or integrator poles), the tuning procedure changes [6]. Although there is a systematic way to cope with this, it is a case by case adjustment. With the KP, the system characteristics do not matter, there will always be only one tuning parameter and no change is required for any particular case whatsoever since the KP inherently copes with any system, as explained in Section 2.

The simulation results for the nominal case (a) and for the uncertain case (b) are presented in Fig. 6. In both simulations there is a reference change at $t = 10$ [min] from 0 to 1, an input disturbance increase of 0.1 is introduced at $t = 130$ [min] and at $t = 250$ [min] noise is added to the output measurement. In the nominal case, as indicated in the previous example, the reference tracking responses are the same, but this does not happen for the disturbance responses, although they are very similar with the exception of KP2, which was tuned for a faster disturbance rejection. In case (b), there is an uncertainty in the delay in both its time-invariant and time-varying parts, which is now given by

$$L(t) = 6.75 + 0.75 \sin(2\pi 0.1t) \rightarrow L \in [6, 7.5],$$

or, as defined in the discrete case in (7),

$$d(k) = 6 + \delta d(k) \rightarrow 0 \leq \delta d(k) \leq 1.5.$$

It can be seen from the results shown in Fig. 6 that all predictors, with the exception of KP2, manage to maintain the system stability even in the face of a time-varying delay. With KP2 and this uncertainty, the closed-loop becomes unstable, showing the

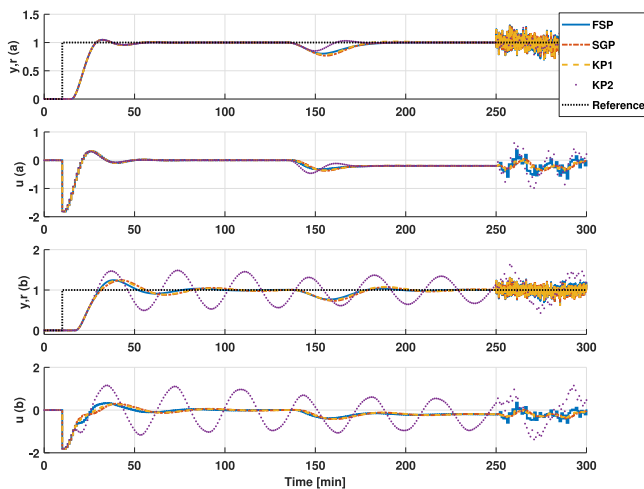


Fig. 6. Example 2: closed-loop responses for (a) nominal and (b) uncertain cases.

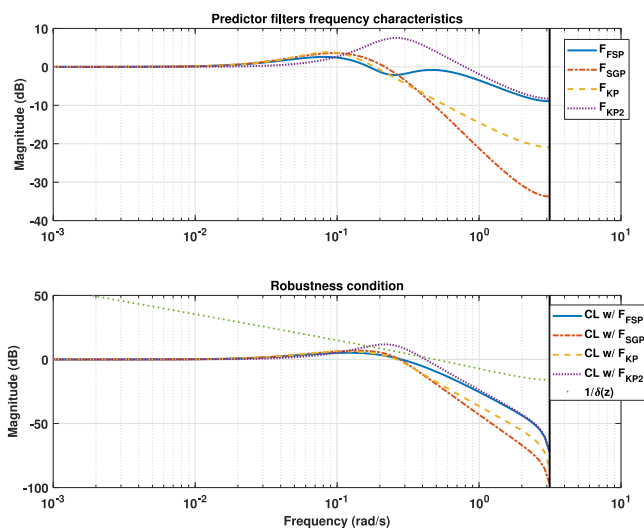


Fig. 7. Example 2. Top: frequency diagram of the predictor filters. Bottom: robustness condition taking into account the different filters and the nominal closed-loop (CL) transfer function.

natural robustness/disturbance rejection trade-off. This stability issue of KP2 should be expected because, when analyzing the robustness condition for all cases through (7), which is shown in Fig. 7, only KP2 do not satisfy the condition.

5. Conclusion

In this work, a novel predictor for dead-time systems based on the Kalman filter was introduced. It was shown that the Kalman Predictor is equivalent to an FSP, which is an important discovery since every property of the FSP and every tool for its analysis can be readily applied to the KP. And, also, this can be done simply by calculating the equivalent prediction-error filter $F_{KP}(z)$.

The proposed KP have some interesting advantages: (i) it can deal with linear systems of any order and dynamics without changes to the tuning procedure; (ii) has only one tuning parameter that can be easily adjusted for disturbance rejection, by increasing its value, or closed-loop robustness, by decreasing its value; (iii) it can properly estimate any kind of disturbance as long as the expected dynamics is included in the predictor during design; (iv) although it considers a fixed delay in its formulation, it can still deal with time-varying delay.

Some simulation examples were presented to corroborate the good behavior and simple tuning of the proposed strategy, showing its relevance even in the face of more complex problems, for example, dealing with sinusoid disturbances and time-varying delays, as the ones dealt in [8,22,23,26]. In future works, the MIMO case where different delays exist for each input will be considered.

CRedit authorship contribution statement

Daniel Martins Lima: Conceptualization, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Supervision, Funding acquisition. **Bruno Martins Lima:** Conceptualization, Formal analysis, Resources, Writing – original draft. **Julio Elias Normey-Rico:** Conceptualization, Formal analysis, Resources, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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